

Spectra Scheduler

Adaptive band selection
from incomplete observations.

Smart Scan strategy for Electronic Warfare

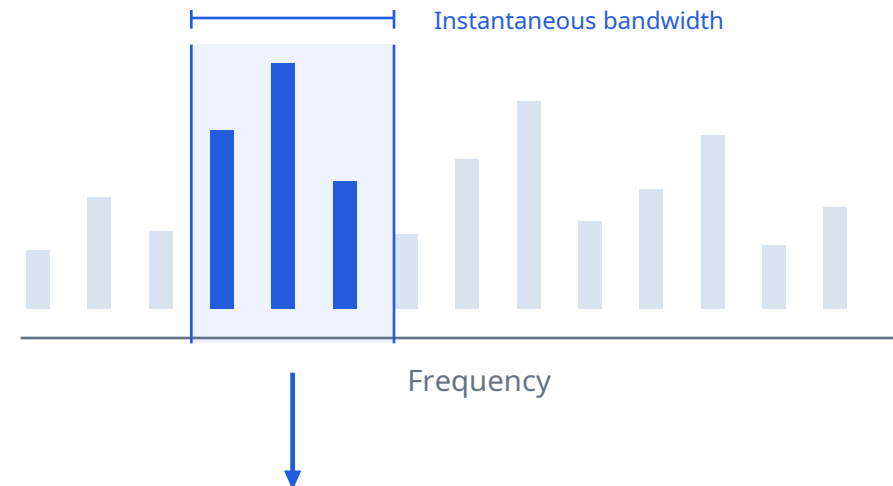
SIH26055 • DRDO • Software

Team name: _____ Team ID: _____

Theme: pending official confirmation

THE BANDWIDTH BOTTLENECK

WIDE SPECTRUM



Limited-bandwidth receiver

Select band / dwell / revisit

**One observation window.
Many signals outside it.**

Listen where evidence and constraints agree.

Proposed solution: jointly manage frequency selection, listening dwell and discovery coverage.

Signals occur elsewhere

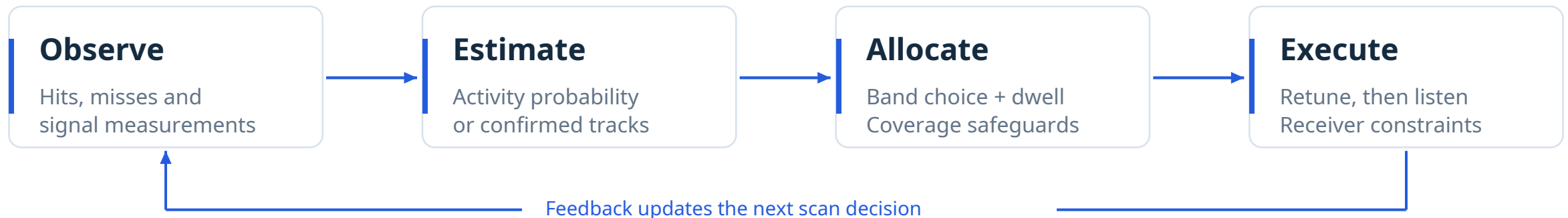
Limited bandwidth creates missed opportunities.

Switching consumes time

Retuning reduces useful listening time.

Activity changes

Old evidence can misdirect the next revisit.



PROPOSED CONTRIBUTION

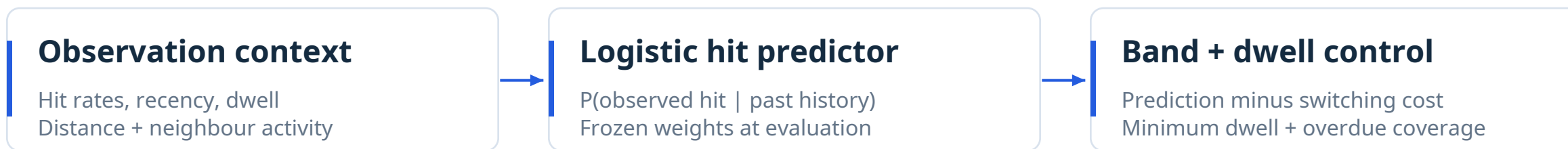
Connect pulse association, predictive revisits and constrained scanning in one evaluated workflow.

Components exist as separate baselines; their unified integration remains planned.

From observed history to a receiver decision.

Python / NumPy • h5py / HDF5 • scikit-learn • JSON model artifacts

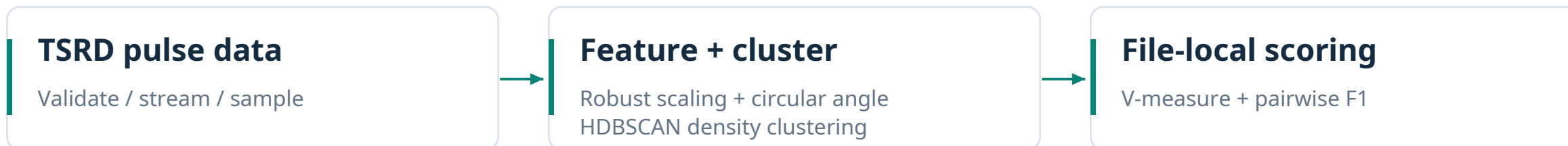
A / IMPLEMENTED ML BASELINE — SIMULATED RECEIVER FEEDBACK



Training target: observed hit / miss. Retuning excluded. No emitter IDs or future truth in model inputs.

Validation: separate seeds, 13 existing strategies and a constant-prediction ablation.

B / IMPLEMENTED OFFLINE ASSOCIATION — TSRD PULSE DESCRIPTOR WORDS



PLANNED: calibrated simulation + association-guided scheduling

Fixed-scan recordings alone cannot evaluate alternative receiver actions.

A working foundation, with clear validation gates.

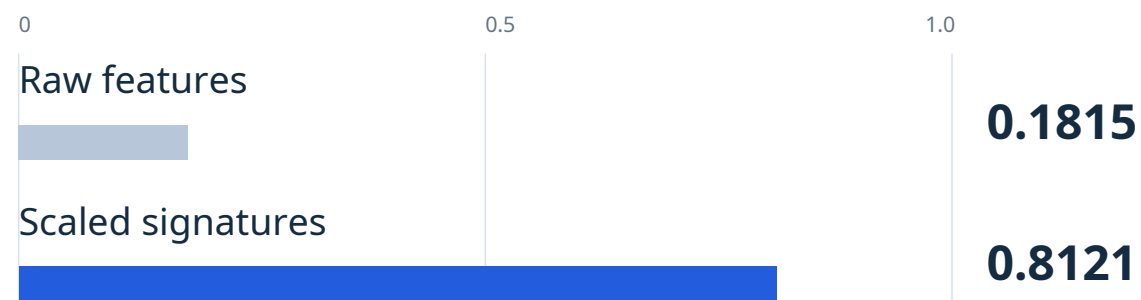
Simulator, streaming data pipeline and model evaluation are operational as software prototypes.

233.2M

scan-training pulses validated

2,500 files • chunked HDF5 processing
Bounded samples; no full-dataset RAM load

ASSOCIATION / PAIRWISE F1



10 training files • 100k sampled pulses • not held-out

False alarms / ambiguity

Track confirmation + expiry
Explicit clustering-noise scoring

Scale / simulation mismatch

Bounded samples + parallel file jobs
Planned unit and behaviour calibration

Predictor may not help

Paired evaluation + model ablation
Retain strong non-ML baselines

CURRENT LIMIT Learned scheduling does not consistently beat the strongest existing policy.

Better observation. A usable evaluation platform.

For receiver-algorithm researchers and spectrum-system evaluation teams; benefits are intended outcomes.

OUTCOME / HOW WE WILL MEASURE IT

Capture more useful activity

Interception and emitter discovery at equal bandwidth.

Respond faster to changing emitters

First-detection and mode-change reacquisition delay.

Use receiver and compute budgets well

Coverage gaps, retuning fraction, memory and latency.

SUCCESS CRITERION

Improve the interception–latency trade-off without unacceptable coverage or resource costs on unseen scenarios.

DELIVERY SCOPE

WORKING

Reproducible experiment engine

Simulator, data pipeline and saved model reports

PLANNED

Integrated scheduling workflow

Dataset calibration + association-informed decisions

PLANNED

Experiment API and dashboard

Run jobs, inspect results and compare configurations

Grounded in research. Tested independently.

References inform the methods; published scores are not presented as project results.

- 01** **Benchmark foundation**  **The Turing Synthetic Radar Dataset**
Gunn et al. / arXiv:2602.03856

- 02** **Data & evaluation contract**  **TSRD dataset card + challenge implementation**
PDW schema, receiver modes, splits and scoring

- 03** **Representation-learning research**  **Radar Pulse Deinterleaving with Transformer**
Based Deep Metric Learning / arXiv:2503.13476

PROJECT EVIDENCE dataset-workflow.md / learning-workflow.md / reproducible JSON reports